Ch. 15 CW #1 Solutions

Principles of Microeconomics

December 16, 2025

15.1. What is a Competitive Market

1. (a) No, there aren't many sellers.
   (b) Yes, natural gas is exactly the same regardless of the producer; there are many buyers of natural gas, and the problem makes it sound like there are many sellers.
   (c) No, the products aren't all the same.
   (d) No, each market has at most one or two ticket sellers, so there aren't many sellers.

2. (c) – In a perfectly competitive market, firms are price takers.

3. (a) Milk is essentially the same regardless of the producer, and there are many buyers and sellers.

   (b)

Quantity	Price	Total Revenue	Average Revenue	Marginal Revenue
1 gallon	\$6	\$6	\$6	\$6
2	6	12	6	6
3	6	18	6	6
4	6	24	6	6
5	6	30	6	6
6	6	36	6	6
7	6	42	6	6
8	6	48	6	6

4. (c) – Marginal revenue is the price of the good which is constant in a perfectly competitive industry. Total revenue is PQ, so it's proportional to the quantity sold. If quantity increases by 10%, total revenue increases by 10%.

15.2. The Competitive Firm's Supply Curve

5. (a) Marginal benefit is the number of deaths due to disease that inoculation prevents. Marginal cost is the number of deaths that inoculation

causes.

MB	MC	Profit
10	0	10
9	1	18
8	2	24
7	3	28
6	4	30
5	5	30
4	5	29
3	5	27
2	5	24
1	5	20

(b) Between 50% and 60% of the population should be inoculated because that saves the highest number of people, 30. Note that it's where $MC = MB$.

(c) Profit is the net number of loves saved – the deaths due to disease prevented minus the deaths due to inoculation. See part (a) for the calculation.

6. In the summer, lobster is locally available and cheaper. For the rest of the year, the shacks have to source lobster from around the world which is more expensive. Sourcing the lobster from around the world increases the shacks' variable costs. They shut down when their average variable costs are higher than the price that they can receive for their lobster. That's optimal because they save more money (their variable cost) than they give up (their revenue).

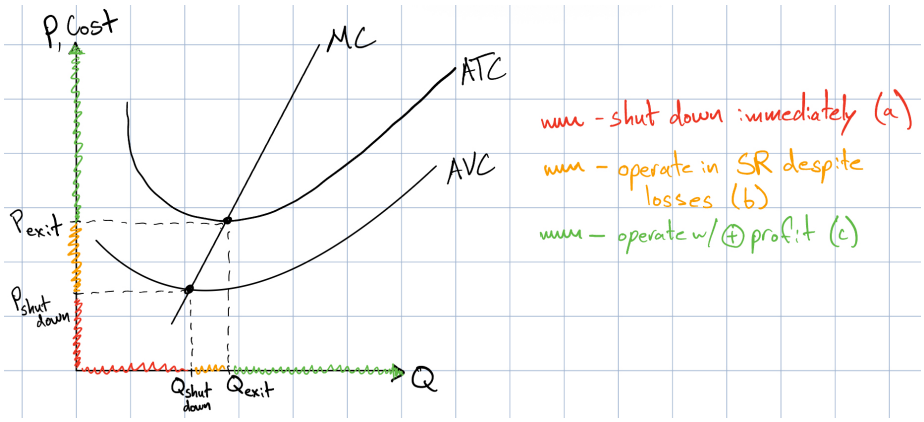
7. (a) MC

   (b) AVC

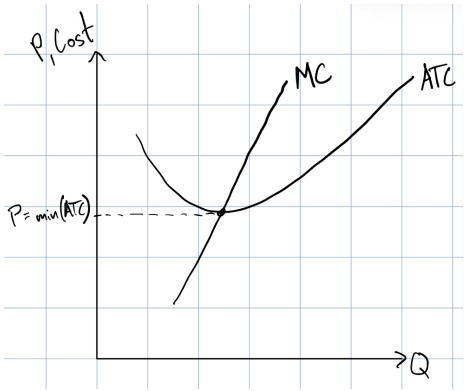
   (c) AVC; ATC

   (d) ATC

8.


P, Cost
MC
ATC
AVC
P_{exit}
$P_{shut\ down}$
Q
$Q_{shut\ down}$
Q_{exit}
www - shut down immediately (a)
www - operate in SR despite losses (b)
www - operate w/ ⊕ profit (c)

9.


P, Cost
MC
ATC
$P = \min(ATC)$
Q